

No. of Functional Features Included in the Solution

Date	03 July 2026
Team ID	N/A
Project Name	OptiCrop – Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Functional Features Overview:

This document lists the functional features implemented in OptiCrop, each with its purpose, implementation status, and contribution to the overall solution.

S. No	Feature Name	Feature Description	Module / Component	Status	Marks Contribution
1	Soil & Climate Input Form	Collects N, P, K, temperature, humidity, pH, and rainfall from the user	Frontend — Find Your Crop page	Done	Core
2	Crop Prediction Engine	Uses the trained Random Forest model to predict the most suitable crop	Backend / ML (app.py)	Done	Core
3	Multi-Algorithm Comparison	Trains and compares KNN, Logistic Regression, Decision Tree, and Random Forest	Model training (train_model.py / notebook)	Done	Core
4	Feature Scaling	Applies StandardScaler to inputs before prediction for consistent results	Backend / ML preprocessing	Done	Core
5	Model Persistence	Saves the trained model and scaler via Pickle for instant reuse without retraining	Backend (model/model.pkl, scaler.pkl)	Done	Core
6	Exploratory Data Analysis	Univariate, bivariate, and multivariate visualizations of the dataset	Notebook (EDA)	Done	Supporting
7	Seasonal Crop Grouping	Groups crops into Summer/Winter/Rainy buckets based on temperature/humidity/rainfall	Data preprocessing	Done	Nice-to-have
8	Multi-page Navigation	Home, About, and Find Your Crop pages with a shared navigation bar	Frontend (templates/)	Done	Core

Feature Summary:

Metric	Count / Value
Total Features Planned	8

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Total Features Implemented	8
Core / Must-Have Features	6
Additional / Nice-to-Have Features	2
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Soil & Climate Input Form; Multi-page Navigation
2	Backend / Logic	3	Crop Prediction Engine; Feature Scaling; Model Persistence
3	Database / Storage	1	File-based storage — crop_data.csv, model.pkl, scaler.pkl (no traditional database)
4	API / Integration	0	None currently integrated (self-contained ML model)
5	Security / Authentication	0	Not applicable — OptiCrop is a public tool with no login required